

6(a)(i)

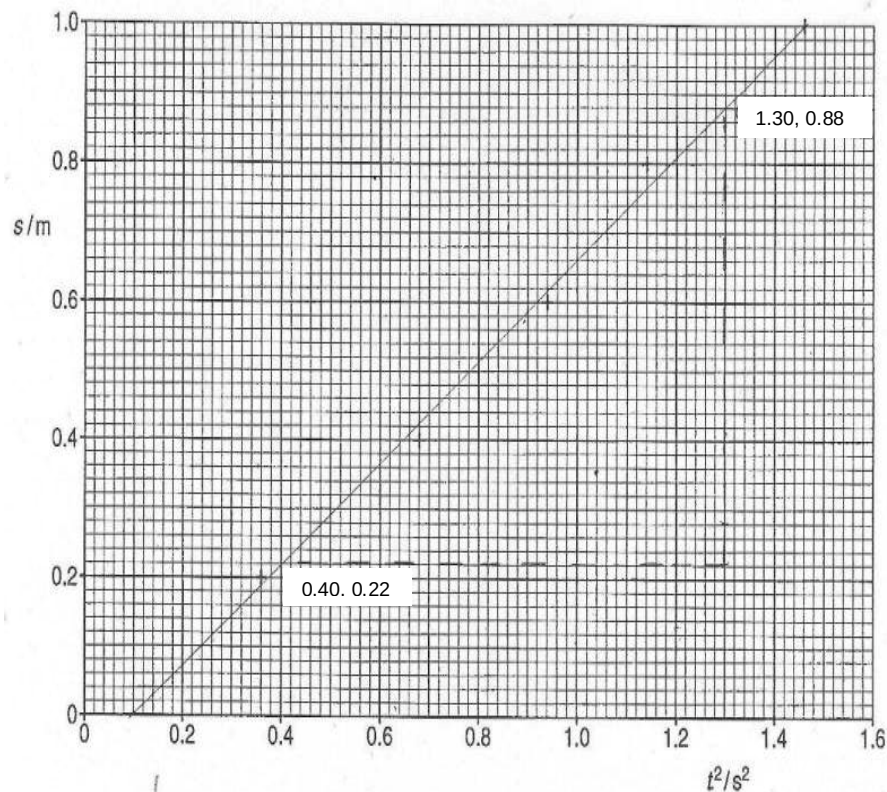


Fig. 6.2

$$s = ut + \frac{1}{2}at^2$$

$$u = 0,$$

$$s = \frac{1}{2}at^2$$

$$\text{Gradient} = \frac{a}{2} = \frac{0.88 - 0.22}{1.30 - 0.40} = 0.733$$

$$a = 2 \times 0.733$$

$$a = 1.5 \text{ m s}^{-2}$$

6(a)(ii) 1

Random error

The line of best fit does not pass through all the five data points as expected as the acceleration is constant. Four of the data points are scattered about the line of best fit.

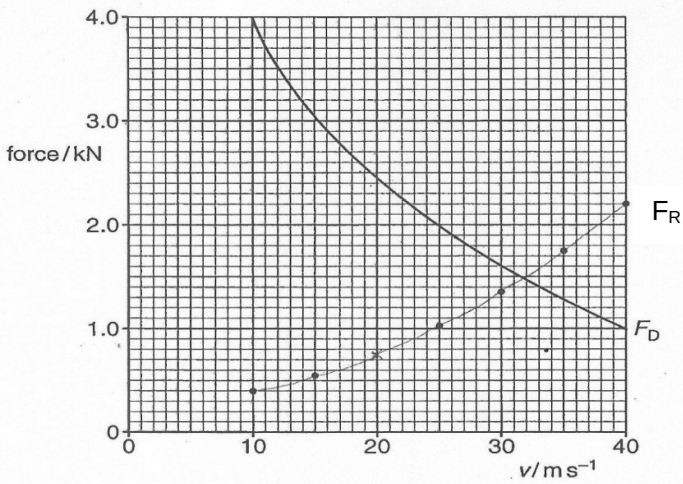
6(a)(ii) 2

Systematic error

The line of best fit does not pass through the origin as expected.

6(a)(iii)

Since line of best fit has two data points scattered on both sides of the line, the errors in the two readings which give higher values than the actual value will partially cancel the errors of the two readings which give lower values than the actual reading. Hence, the effect of random error is reduced.

6(b)(i)	$F_R = 280 + 1.2(20)^2$ $= 760 \text{ N}$
6(b)(ii)	 The graph shows force in kN on the y-axis (0 to 4.0) and velocity in m s⁻¹ on the x-axis (0 to 40). A curve labeled F_D (drag force) starts at (0, 4.0) and decreases. A series of points connected by a line labeled F_R (resistive force) starts at (10, 0.4) and increases. The two curves intersect at approximately (31.8, 1.48). Fig. 6.3
6(b)(iii)1	Maximum speed occurs at the point of intersection of the two graphs F_D and F_R. According to Newton's Second Law of Motion, $F_{\text{net}} = ma$. If $F_{\text{net}} = 0$, $a = 0$ and $v = \text{constant}$. Hence $F_D = F_R$ From the graph, maximum speed = 31.8 m s^{-1} (read to 1/5 of a small square)
6(b)(iii)2	Total power output $P = F_D \cdot v$ $F_D = 1.48 \text{ kN}$ $P = (1.48)(10^3)(31.8)$ $= 4.71 \times 10^4 \text{ W}$
6(b)(iii)3	From Fig. 6.3, when $v = 20 \text{ m s}^{-1}$, $F_R = 0.75 \text{ kN}$ and $F_D = 2.45 \text{ kN}$. $F_{\text{net}} = F_D - F_R$ $ma = 2.45 - 0.75$ $950a = 1.70 \text{ kN}$ $a = 1.79 \text{ m s}^{-2}$
6(c)	When the car is travelling up a slope, there is a component of its weight acting downwards on the car. Hence the final resistive force F_R' is now larger than the initial resistive force F_R. The F_R' graph will now intersect the F_D graph at a lower maximum speed.
7(a)	The origin of upthrust is due to the difference in fluid pressure between the